Muscle Cola demand exercise

This uses the 46 responses in muscle-cola.csv to build a demand curve for one cola sold through gyms. I checked the results the assignment gave us before using the curve for the other prices.

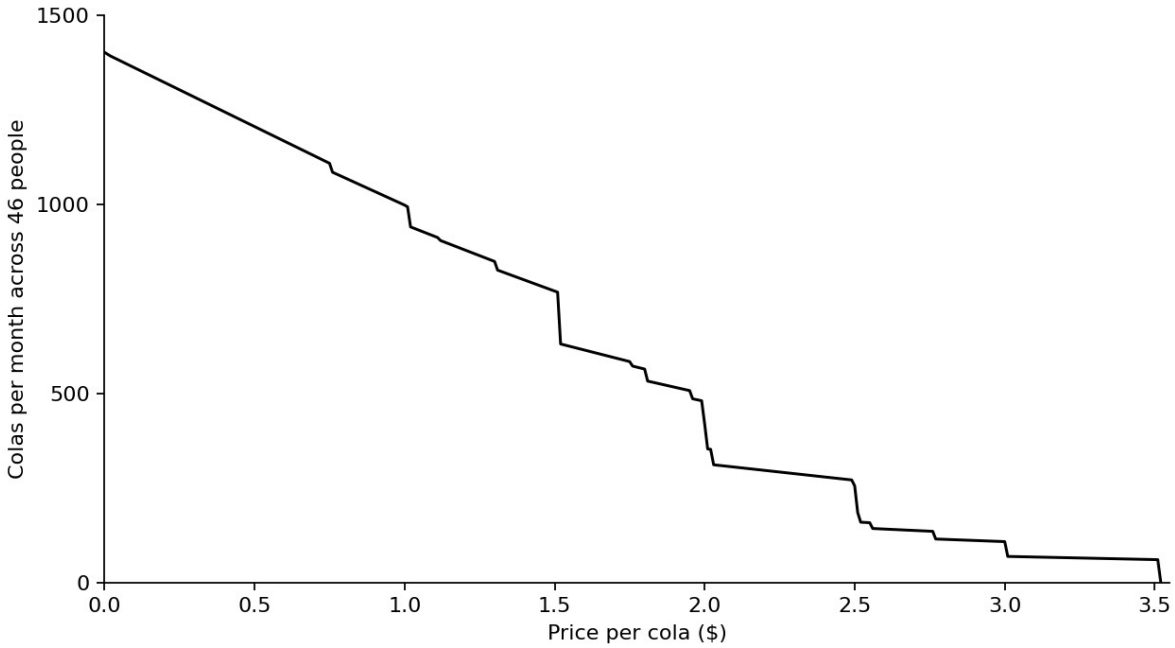
Part 1 Check the file and calculation

Check	My result	Given answer
Number of respondents	46	46
WTP equal to $0	11	11
Would take none even if free	12	12
Highest WTP	$3.51	$3.51
Median WTP among people willing to pay	$1.99	$1.99
Quantity at $1.00	997.90	about 998
Quantity at $2.00	418.98	about 419

All seven matched. I included every respondent, including people who answered zero.

Part 2 Demand curve

I drew a straight line for each person from the number they would take for free to the number they would take at their maximum price. Above that maximum, their quantity is zero. Adding the 46 lines gives the curve below. The quantities between each person’s two answers are interpolated along that line.



The quantities I read off are:

Price	Colas per month across the 46 people	Method
$0.00	1,402.00	Counting
$0.50	1,205.62	Counting
$1.00	997.90	Counting
$1.50	770.45	Counting
$2.00	418.98	Counting
$2.50	255.04	Counting
$2.51	184.51	Counting
$3.00	107.72	Counting
$3.51	60.00	Counting
$3.52	0.00	Counting

By counting, I mean summing the individual lines at each price. I did not fit a separate statistical curve. At $0, the total is 1,402 colas per month: the sum of everyone's quantity_at_P0 answer, including the twelve zeros.

Part 3 What I noticed

The big drop right after $2.50 stood out to me. Four people said $2.50 was the most they would pay. At that price they would buy 4, 4, 60, and 2 colas a month, so 70 altogether. At $2.51 they all drop out under the rule for this exercise, and total quantity goes from about 255 to 185. That means a profit estimate near $2.50 could change a lot based on a tiny price difference, especially because one person makes up most of that drop. Revenue for these 46 people would go from about $638 to $463 a month. I cannot tell what happens to profit without knowing the cost of each cola.

Part 4 AI log

I first gave the assignment to the AI, but it could not find the CSV. It stopped and asked for the file instead of guessing. That was the dead end. After I attached muscle-cola.csv, it checked all 46 rows and returned the five file checks and the quantities at $1 and $2. I compared those seven numbers to the answers in the assignment. They all matched before I used its other results.

The AI then calculated the other prices, made the chart, and helped me notice the four respondents at $2.50. There was no incorrect number to fix after I provided the file. I kept the zero answers and did not call the revenue example a profit estimate.